

*International networking on in vitro colon models
simulating gut microbiota mediated interactions*



Action CA23110

Data management and FAIR principles

INFOGUT 3rd Training school

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Two minutes. Answer them in your head. Nobody will be asked for an individual answer.



Could you re-run it?

Take the last figure you published or presented. Starting from the raw files, could you reproduce that exact number today?

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Where is it written?

A reviewer asks which samples you excluded, and why. Where is that recorded, and could you find it this afternoon?

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Could someone take over?

If you left tomorrow, how long would a colleague need to pick up your project and carry on with it?



Discuss at your table

Which of the questions was the most uncomfortable to answer.

Discuss with the others why these questions are hard to answer.

Pick one answer that worried the table most and choose someone to give it back to the room in a single sentence.

5

MINUTES

No notes, no slides, no solutions yet. Just compare.



What the “reproducibility” crisis suggests

Fabricated numbers. Duplicated gel images. Deliberate misconduct. Real, investigated, and not what this session is about.



What it actually looks like

- A threshold chosen once and never written down.
- An intermediate file overwritten by the next version of itself.
- A sample dropped for a perfectly good reason that nobody recorded.
- Knowledge that lives in one head, and leaves when that person does.

Which is good news: the fix is boring, and you can start it this week.



RUN 1

A clean result

A postdoc runs the colon model: one control, two treatments, technical replicates, community composition at endpoint. Treatment A produces a strong and convincing shift. The work is written up.



RUN 2, TWO YEARS LATER

Nothing happens

A new PhD student repeats it as the starting point of her own project. Same lab. Same protocol document. Same inoculum source. Treatment A does almost nothing.

Nobody did anything wrong. So, what could have changed?



At the bench



At the keyboard



At the bench

- Medium batch and supplier
- Inoculum donor, storage and thawing
- Vessel, working volume, pH control
- Incubation and sampling times
- Extraction kit and lot number
- Primer set and library preparation



At the keyboard

- Sequencing run and depth
- Pipeline and software versions
- Reference database release
- Filtering and normalisation thresholds
- Which replicates were excluded
- Defaults that changed with an update



In this session

The data management cycle. FAIR principles. Minimum information standards. Organising and documenting a dataset so that a stranger can use it.



Not in this session

Code reproducibility: environments, dependencies, package versions, random seeds.



What you leave with

A small dataset you have restructured yourself, with its metadata file and its README written, organised and ready to push in the GitHub session.

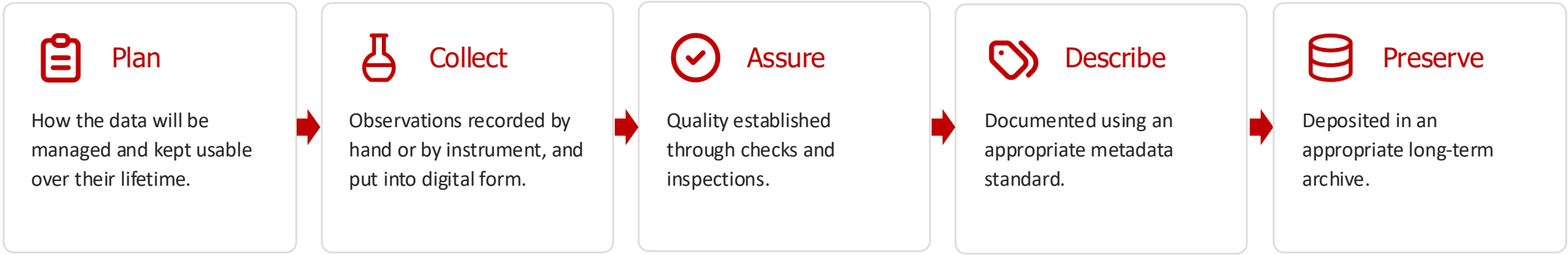


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Data life-cycle, DMP and FAIR

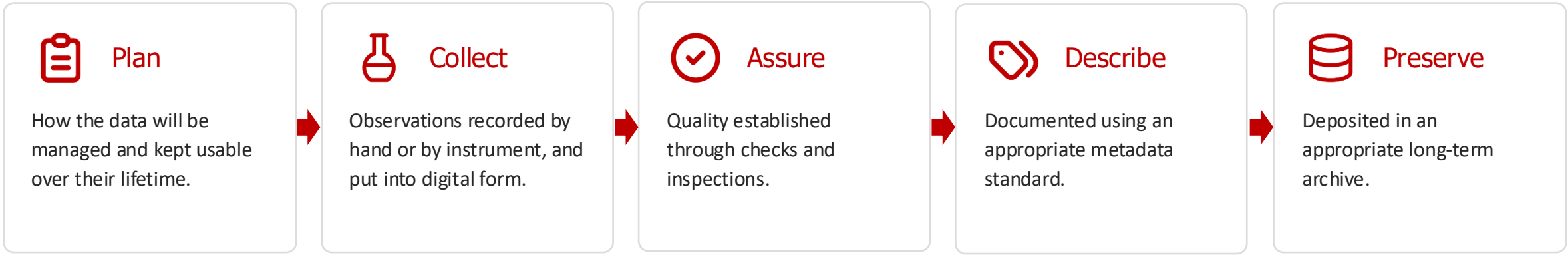
Concepts and definitions

WHEN YOU GENERATE THE DATA YOURSELF



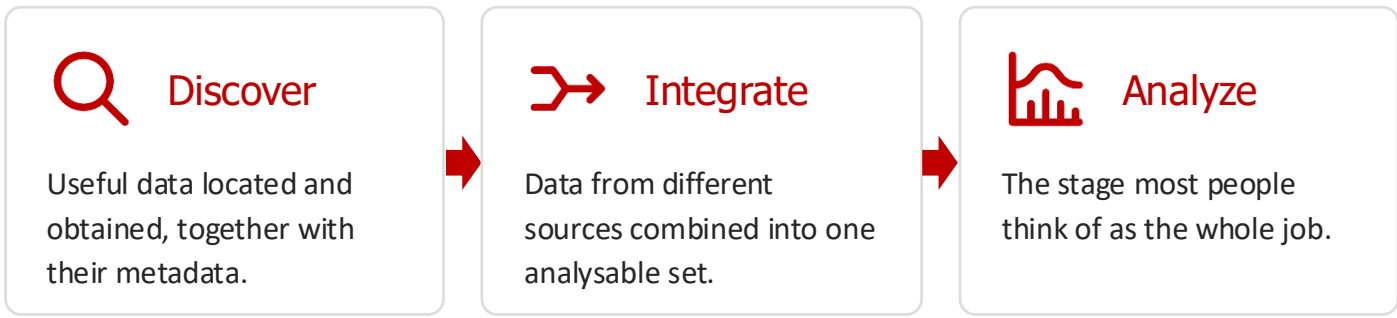
Assure and Describe are stages of their own, not things you do afterwards if there is time.

WHEN YOU GENERATE THE DATA YOURSELF



Assure and Describe are stages of their own, not things you do afterwards if there is time.

WHEN YOU REUSE DATA THAT ALREADY EXIST



Source: DataONE Data Life Cycle

The same failure, recording, repeated at four different stages.



COLLECT

Medium batch and supplier, inoculum handling, extraction kit lot. Some of it in a notebook, some of it nowhere.



ASSURE

One replicate looked odd and was quietly dropped. The decision was made at the screen and never written down.



DESCRIBE

What treatments A and B actually were. Obvious to the person who ran it, opaque to everyone who came after.



PRESERVE

The processed tables lived on a laptop that has since been wiped, and the person who owned it has moved institution.

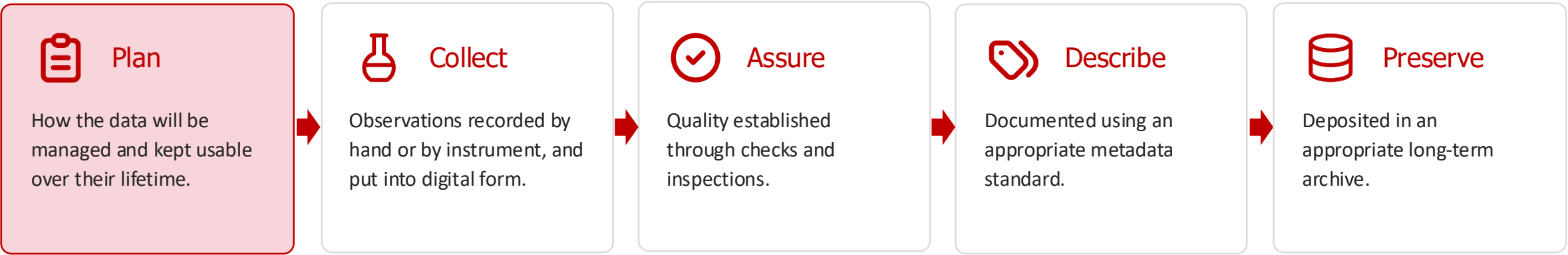


At your table, five minutes

Which of these four would your own lab fail first? Pick one, and say what you would have to change to stop failing it.

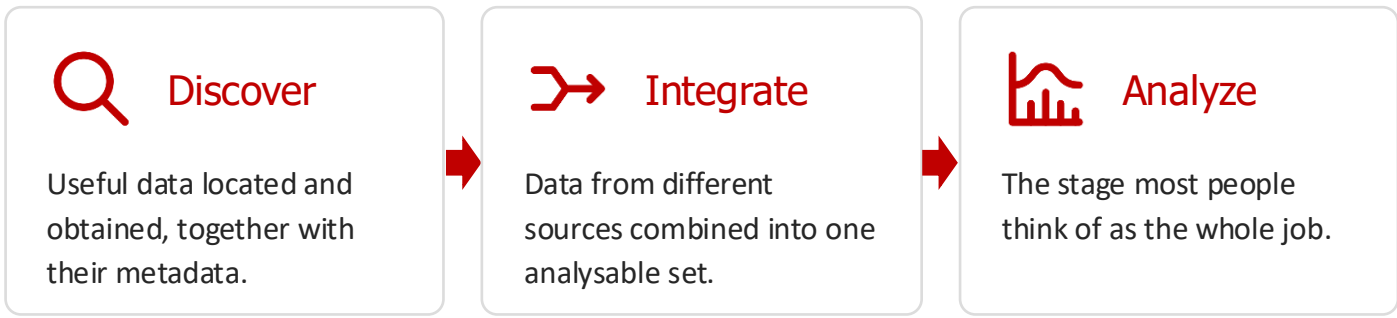
Every one of these is cheaper to fix upstream than to reconstruct later.

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Source: DataONE Data Life Cycle

The first stage of the cycle has a name, a template and a deadline.



What it is

A short document written at the start of a project. What data will be produced, in what formats, where they will live, who is responsible for them, and what will eventually be shared.



Why funders ask

Public money buys a reusable asset, not only a paper. Managing data along FAIR principles and giving open access to them, as open as possible and as closed as necessary, is a mandatory practice for most funders.

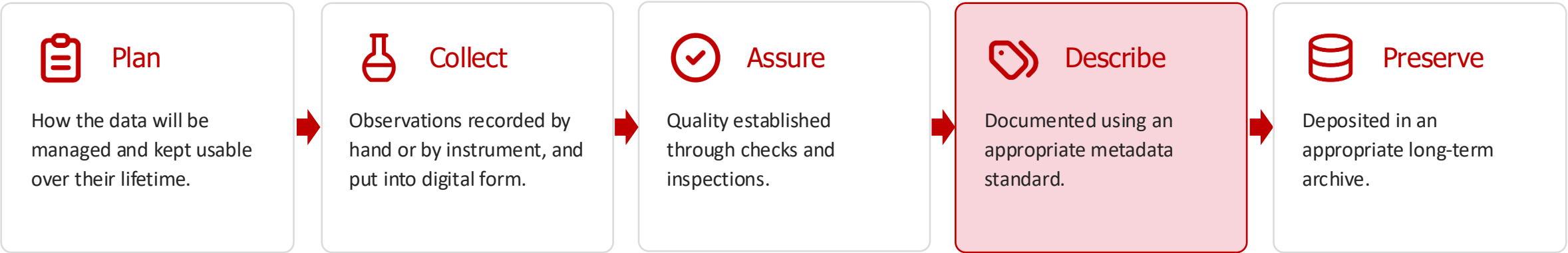


Why it helps you

It forces three decisions : where the data live, who owns them, and what gets shared.
It is also the only way storage and repository costs end up in the budget, since unbudgeted costs cannot be charged later.

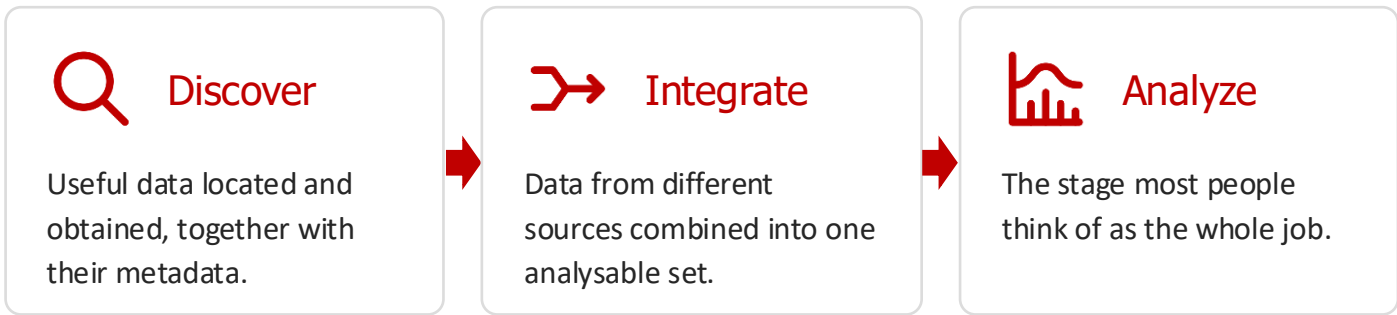
Write one properly once, and the structure carries over to every project after it.

WHEN YOU GENERATE THE DATA YOURSELF



Assure and Describe are stages of their own, not things you do afterwards if there is time.

WHEN YOU REUSE DATA THAT ALREADY EXIST



Source: DataONE Data Life Cycle

Asked about your dataset by somebody you will never meet.



F — FINDABLE

Can they find it?

It exists somewhere public, it carries a persistent identifier, and it is described well enough to turn up in a search that was not looking for it by name.



A — ACCESSIBLE

Can they get it?

Retrievable through that identifier by a standard protocol. Open where possible. Where it cannot be open, the conditions for access are stated not implied.



I — INTEROPERABLE

Can their tools read it?

Open formats, standard vocabularies, explicit units. A plain CSV not a colour-coded workbook that only means something on your screen.



R — REUSABLE

May they use it, and do they know enough to?

Clear provenance, metadata rich enough to judge whether it fits their question, and a licence saying what is allowed.

FAIR is not a synonym for open. A restricted dataset can be FAIR, provided the conditions are stated.

Three of the four FAIR questions are about software, not about people.



What you can read

Sample	OD
Control R1	0.42
A R1	0.38
A R2	n/a

The pink row means contaminated. That fact exists only in your head and in this one screenshot. No script can filter on a colour, and "n/a" is not a number.



What a machine can read

sample_id	od600	qc_flag
Control_R1	0.42	pass
A_R1	0.38	contaminated
A_R2	NA	missing

The same information, in a column a script can filter on. The unit is in the column name. Missing values use one agreed code, every time.

Not bureaucracy. A prompt for the things you would otherwise forget to record.



MIxS

GENOMIC STANDARDS CONSORTIUM

The metadata backbone. Separate checklists for genomes, metagenomes, marker gene sequences, metagenome-assembled genomes and uncultivated viral genomes, plus environmental packages describing the habitat a sample came from. The foundation for standardised microbiome metadata since 2011.



STORMS

NATURE MEDICINE, 2021

Seventeen items across the six sections of a paper, built on MIxS together with the STROBE and STREGA epidemiology guidelines. Scoped to human microbiome research, so not your study design, but worth knowing it exists.



STREAMS

NATURE MICROBIOLOGY, 2025

Expands STORMS to 67 items covering environmental, synthetic and non-human host-associated studies, explicitly including engineered systems. Machine-actionable, aligned with existing metadata standards, and built from the input of 248 researchers across 28 countries.

Sources: Yilmaz et al. 2011; Mirzayi et al. 2021; Kelliher et al. 2025

A checklist can only be as specific as the community that wrote it.



Record it anyway

Vessel type and working volume

Medium composition, batch and supplier

Inoculum provenance, storage and thawing

pH control, temperature, retention time

Feeding and sampling schedule

How anaerobic conditions were maintained



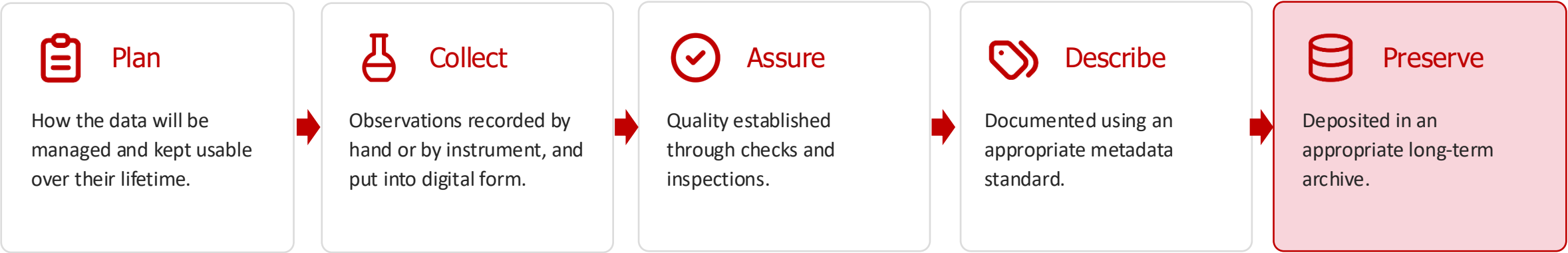
Why that falls to you

If the fields your experiment needs do not exist in any checklist yet, that is not permission to skip them. Put them in your own metadata file, with clear names, explicit units and stated allowed values.

A well-documented supplementary file is what a standard looks like before it is a standard. The fields you and your colleagues converge on are the evidence the next checklist will be built from.

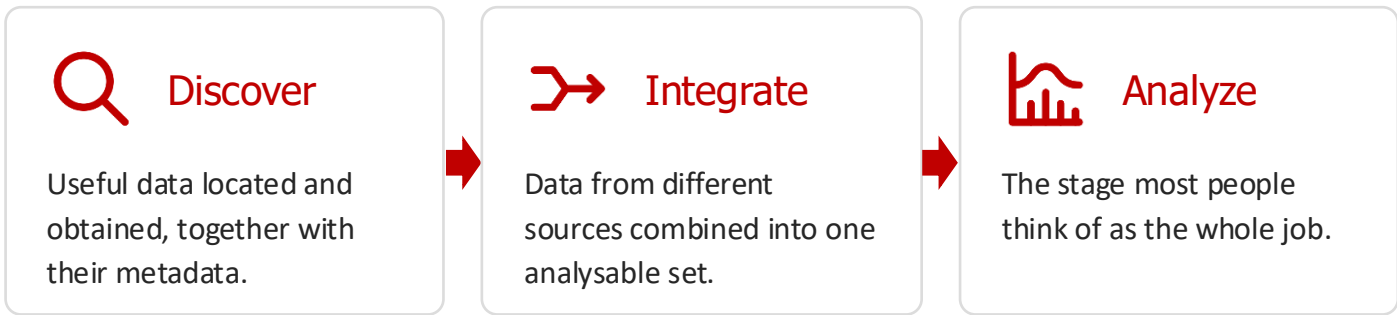
You are not waiting for this standard to arrive. You are the people it will come from.

WHEN YOU GENERATE THE DATA YOURSELF



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WHEN YOU REUSE DATA THAT ALREADY EXIST



Source: DataONE Data Life Cycle

Deposit is not the end of the project. It is the thing that makes the rest of it citable.



ENA

Nucleotide sequence data: raw reads, assemblies and their sample metadata. Hosted at EMBL-EBI.



SRA

The counterpart at NCBI for raw sequencing reads, widely used by North American collaborators and journals.



MetaboLights

Metabolomics experiments, raw and derived data, together with the study metadata that go with them.



Zenodo

Everything that has no domain repository: tables, documents, figures, supplementary data. Issues a DOI.

Every deposit hands you back an accession number or a DOI.

That string belongs in your paper, in your README and in your metadata file. It is the persistent identifier that makes the Fin FAIR true, and it is the only permanent address your data will ever have.



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Data horror-show

Let's assess bad practices... and curate our own dataset!

EXHIBIT 1

 **scripts/** — **folder contents**

```
analysis_final.R  
analysis_final_v2.R  
analysis_final_USE_THIS.R  
analyse2_old.R  
analysis_final_v2 (1).R
```

EXHIBIT 1

 scripts/ — folder contents

```
analysis_final.R
analysis_final_v2.R
analysis_final_USE_THIS.R
analyse2_old.R
analysis_final_v2 (1).R
```

EXHIBIT 2

 project/ — folder contents

```
Control_R1.csv
script.R
plot1.png
notes.txt
data2.csv
final_table.csv
... 34 more files
```

EXHIBIT 1

 scripts/ — folder contents

analysis_final.R
analysis_final_v2.R
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EXHIBIT 2

 project/ — folder contents

Control_R1.csv
script.R
plot1.png
notes.txt
data2.csv
final_table.csv
... 34 more files

EXHIBIT 3

 metadata.xlsx — spreadsheet

	A	B	C
1	Microbial Ecology Lab		
2			
3	Experiment 4 FINAL		
4			
5	Run by JD, Mar 2024		
6			
7	Sample	OD	pH
8	Control R1	0.42	6.8

EXHIBIT 4

 README.md — file contents

```
# Project
```

```
scripts for the paper
```

EXHIBIT 4

 README.md — file contents

```
# Project

scripts for the paper
```

EXHIBIT 5

 metadata.csv — table

sample_id	od600
S1	0.42
S2	0.51
tube 4 (new)	0.38
tube 4 (new) redo	0.39
CTRL old	0.44
ctrl_2	0.45

EXHIBIT 4

 README.md — file contents

```
# Project

scripts for the paper
```

EXHIBIT 5

 metadata.csv — table

sample_id	od600
S1	0.42
S2	0.51
tube 4 (new)	0.38
tube 4 (new) redo	0.39
CTRL old	0.44
ctrl_2	0.45

EXHIBIT 6

 metadata.csv — table

sample_id	date_collected
Control_R1	01/02/2024
Control_R2	02/01/2024
A_R1	Feb 1 2024
A_R2	1 Feb 24
B_R1	2024-02-01
B_R2	44958



Every one of those is information that exists in exactly one person's head.



Nobody meant to

Each of those was the fastest thing to do on the day it happened. None of them was a mistake at the time, which is exactly why they are so common.



You cannot see it from inside

While you still remember, every one of those files is perfectly clear. The problem only appears when the memory fades, or when somebody else opens the folder.



Fix it while you remember

Not at submission, not when a reviewer asks, not when you hand over. Now, while you can still say what every file is and why it is there.

Why a metadata file, when the filename already says it

Control_R1 already tells you it is a control and a first replicate. So why write it down again?



Two different readers

A filename is readable by a human who was there. A metadata file is readable by a script and by a stranger.

Filtering on a column works in any language. Parsing a filename works until somebody adds Control_R1_repeat.

And no filename will ever tell a reader what treatments A and B actually were.



data/metadata.csv, six rows

```
sample_id,condition,replicate,  
replicate_type,timepoint,file
```

```
Control_R1,control,R1,  
technical,endpoint,...  
A_R1,A,R1,technical,endpoint,...
```

Before you start : what other columns would this file need if these were your own experiments?

Raw data is read-only

One habit, one folder layout, and everything downstream stays recoverable.



The layout you will use today

```
project/  
  README.md  
  data/  
    raw/  
    processed/  
  scripts/  
  results/
```



Three rules

Nothing in data/raw is ever edited. You read from it, you never write to it. If it needs fixing, the fix is a script that writes somewhere else.

Everything in results/ can be deleted and regenerated by running what is in scripts/. If that is not true, something is only in your head.

If you cannot tell which files are raw and which are derived, then all of them are effectively raw and none of them are reproducible.

This is the structure you will build in the exercise, and the one you will push in the GitHub hands-on session.

Your turn: clean up what you built this week



Restructure

Move what you have into data/raw, data/processed, scripts and results. Nothing in data/raw gets edited from here on.



Rename

One convention, applied to every file. No spaces, no slashes, no accents.



data/metadata.csv

One row per sample, one column per variable. Explicit units, one agreed code for missing values, no colour and no free text in a numeric column.



README.md

What this is. Where the data came from. What each folder holds. How to run it. Who to ask. Five headings, a few lines each.



Score your folder against the four FAIR questions now, and again when you finish.

Findable, Accessible, Interoperable, Reusable. Yes, partly or no for each. Nobody has to share their first score.

Perfect is not the target. Better than it was an hour ago is the target.



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Wrapping up

Conclusions and links to go further



FAIR is a direction, not a pass mark. A folder that is better than it was this morning is a success.

The only test that matters

Could a competent stranger, holding only your files, carry on?
Everything today was a way of answering yes.



Github session

You now have a folder worth pushing: structured, named, described and documented. GitHub is where it goes next.

Thank you. Questions before we hand over?

Where to look next



GO FAIR

go-fair.org/fair-principles

The fifteen principles sitting behind the four questions, each one explained in a paragraph.



FAIRsharing

fairsharing.org

A curated registry of standards, databases and funder or journal data policies. Search it before you invent a format of your own.



MIXS

gensc.org/mixs

The Genomic Standards Consortium checklists and environmental packages for sequence metadata.



STREAMS

doi.org/10.1038/s41564-025-02186-2

The 67-item reporting checklist for environmental, synthetic and non-human host-associated microbiome studies.

And the research data team at your own institution or attached to the university libraries...